Decision Trees and Pegasos SVMs from Scratch

Price and price-tier prediction on bina.az listings

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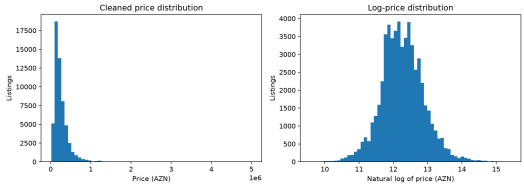
MLE-AI-201 Machine Learning — AI Academy, Cohort II

Final project demo

Problem and dataset

bina.az sale listings (Kaggle dump)

- 100,775 raw rows, 51 columns, Azerbaijani names, text numbers
- After cleaning: **57,363** Baku residential listings, 65 features
- Split 70/15/15, seed 42: 40,155 / 8,604 / 8,604



Heavy right tail: a few listings cost up to 5M AZN; log(price) is near-symmetric.

Two tasks, one dataset

- **Task A** — regression: price in AZN
  RMSE, MAE, R^2
- **Task B** — premium vs standard threshold = train median **209,000 AZN**
  precision, recall, F1, ROC-AUC

Cleaning, features and leakage

Fixed rules, no data statistics

1. Dedupe listing URLs: 100,775 → 64,454
2. Residential + Baku only: 58,257
3. Parse *Sahə*, *Mərtəbə* “7 / 9”, land; yes/no → 0/1
4. Structural fills; drop missing essentials: 57,587
5. Domain outliers (price 10k–5M, area 15–1000 m², …): **57,363**

Features (encoder fitted on train only)
14 numeric + distance to centre + floor ratio + 3 category + 45 districts + `loc_other`

Removed: target leakage

`unit_price` = price / m²
`total_price` = price (99.99%)

Removed: not available / not general

`views`, `updated` (post-listing)
owner & shop names, URLs, address, IDs

Tier threshold, district vocabulary and scaler: training split only.

Our decision tree (NumPy)

Split search

- Sort each feature; thresholds between distinct neighbours
- Prefix counts / sums $\Rightarrow$ every threshold in $O(n)$
- Pick the largest gain $\Delta I = I(\text{parent}) - \sum_c \frac{N_c}{N} I(c)$

Impurities

Gini	$1 - \sum_k p_k^2$
Entropy	$-\sum_k p_k \log_2 p_k$
MSE (regression)	$\frac{1}{N}\sum_i (y_i - \bar{y})^2$

Stopping controls (each tested)

- `max_depth`, `min_samples_split`
- `min_samples_leaf`, `min_impurity_decrease`

Leaves: majority class / mean; `predict_proba` = leaf frequencies

Checks: matches sklearn's splits on a tiny, fully determined example; impurity-decrease importances

Our SVM: Pegasos sub-gradient + random Fourier features

Objective (hinge primal, no QP solver)

$$J(\mathbf{w}, b) = \tfrac{\lambda}{2}\|\mathbf{w}\|^2 + \tfrac{1}{n}\sum_i \max\big(0, 1 - y_i(\mathbf{w}^\top \mathbf{x}_i + b)\big)$$

Update for a random row i, $\eta_t = \frac{1}{\lambda t}$ or constant:

$$\mathbf{w} \leftarrow (1 - \eta_t \lambda)\mathbf{w} + \eta_t y_i \mathbf{x}_i \cdot \mathbf{1}[y_i f(\mathbf{x}_i) < 1]$$

Unregularised bias, projection onto $\|\mathbf{w}\| \le 1/\sqrt{\lambda}$, standardised inputs.

Diagnostics

- Objective logged at 51 checkpoints
- $\mathbf{w}$, margin width $2/\|\mathbf{w}\|$
- Support vectors: $y_i f(\mathbf{x}_i) \le 1$

Non-linearity: RFF

$$z(\mathbf{x}) = \sqrt{2/D}\cos(\mathbf{x}W + \mathbf{c}),\ W \sim \mathcal{N}(0, 2\gamma)$$

Same linear solver; fixed cost D instead of $O(n)$ kernel sums per step.

Experimental setup

Selection on validation only

- Tree: depth $\{2, 4, 8, 12\}\times$ leaf $\{1, 5, 20\}$, Gini vs entropy
- SVM: 33 predeclared configs (λ, schedule, T, γ, D)
- Ridge (bonus): $\alpha \in \{0.1, \ldots, 1000\}$
- Test split scored **once** with frozen settings

Uncertainty

5-fold CV on train+val (test excluded), refitting encoder, scaler and tier threshold per fold; mean $\pm$ std

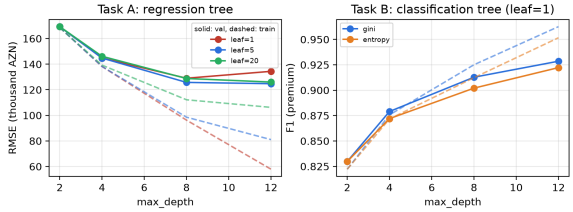
Matched sklearn baselines

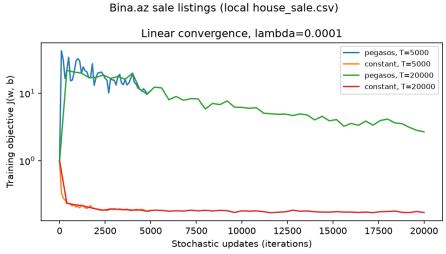
- sklearn decision trees: same grid and criteria
- `SVC` linear / RBF in a `Pipeline` with `StandardScaler`
- Our loss averages the hinge, SVC sums it: $C = 1/(n_{\text{train}}\lambda)$

Reproducibility

`python -m src.run_all` rebuilds every number and figure; seeds fixed, deps pinned, data and source hashes in the manifest; 348 tests pass

Tuning: bias–variance and convergence





- Leaf 1: validation RMSE **rises** after depth 8 $\Rightarrow$ overfitting
- Leaf 5 + depth 12 selected (val RMSE 124.7k)
- Gini $>$ entropy at every depth $\geq$4 (F1 0.929)

- Constant η: objective 1.0 $\rightarrow$ 0.31
- $1/(\lambda t)$ at $\lambda{=}10^{-4}$: huge first steps, not converged after 20k updates
- Larger λ: wider margin (0.45 $\rightarrow$ 0.71) but lower F1 (0.887 $\rightarrow$ 0.871)

Results (test split, 5-fold CV mean $\pm$ std)

Task A — price (AZN)

Model	RMSE	R^2	CV RMSE
Ours: tree	**109,164**	**0.760**	119.7k $\pm$ 3.0k
sklearn tree	119,376	0.712	123.5k $\pm$ 3.0k
Ours: Ridge	128,441	0.667	136.8k $\pm$ 5.5k
sklearn Ridge	128,441	0.667	136.8k $\pm$ 5.5k

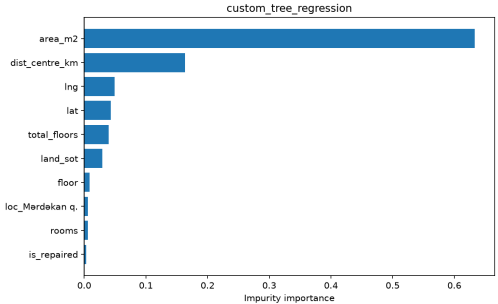
sklearn's search picked depth 8, ours depth 12. At matched settings the trees agree; Ridge matches to numerical precision.

Task B — premium tier

Model	F1	CV F1	Fit (s)
Ours: tree	**0.929**	0.925 $\pm$ .003	2.8
sklearn tree	0.929	0.925 $\pm$ .003	0.3
Ours: linear SVM	0.881	0.885 $\pm$ .005	**0.3**
Ours: RFF SVM	0.846	0.836 $\pm$ .009	1.2
SVC linear	0.903	0.906 $\pm$ .002	43.1
SVC RBF	0.893	0.893 $\pm$ .001	54.2

SVC solves the same objective exactly; our 20k updates are half an epoch, but train 140$\times$ faster.

What the models learn and where they fail



Tree importance: area 0.63, distance to centre 0.16. SVM weights agree: area +3.6, house −1.0, distance −0.85.

Luxury outliers

- Above train P90 (455k AZN): MAE **177k** vs 31k for the rest; underpredicted
- Worst 1% of rows ⇒ 70–74% of squared error
- Villas in Bilgəh, Badamdar, Səbail: view, finish, exact address are not in the data

Sparse districts (`loc_other`)

- MAE 54.7k vs 44.7k; SVM error rate 16% vs 12%

Tier errors sit near the 209k threshold only 7 of 856 expensive listings misclassified by the tree

After the benchmark: improvements on validation only

SVM: optimisation, not the model

- 20 epochs + iterate averaging
- Linear F1 0.887 $\rightarrow$ **0.907** (SVC: 0.908)
- RFF ($\lambda{=}10^{-5}$) F1 0.841 $\rightarrow$ **0.898**

Price: bagging our own trees (bonus)

- 48 trees, depth 32, target $\log(\text{price}/\text{area})$
- Val RMSE 124.7k $\rightarrow$ **97.2k**, MAE 48.3k $\rightarrow$ 34.0k, R^2 0.82
- Val MAE keeps falling as trees are added

Honest limitations

- Asking prices, Baku residential only
- Test split already seen $\Rightarrow$ improvements reported on validation, not test
- CV = variability of frozen settings, not nested selection
- 446 exact duplicate feature rows remain (effect $<$1% MAE)
- Single-run timings on one laptop

Who did what and next steps

Member	Main contribution
Davud Vahidov	Cleaning, encoding, leakage audit; quality & price studies
Sanan Hajiyev	Decision tree, sweeps, sklearn parity tests
Farid Samedov	Pegasos SVM, RFF, 33-config SVM study
Sabuhi Khamiyev	Metrics, evaluation, plots, subgroup errors
Bailar Bayramov	Ridge bonus, `run_all`, integration tests

Next

- Location & quality features (address, view, renovation)
- Deduplicate by content, not URL
- Fresh untouched test set for the ensemble

Code: `github.com/EmbeddedDesignEngineer/ML-Final` (tag `v1.0-final`)